Human influences on ecosystems

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

1 · Part 1

Core · Food supply · [1]

Answer: B

Explanation

Herbicides kill weeds, so the crop plants face less competition for light, water and mineral ions. A describes the effect of fertiliser, C the effect of insecticide and D the effect of irrigation; a herbicide does none of these.

2 · Part 1

Core · Food supply · [1]

Answer: D

Explanation

A monoculture concentrates one crop and can allow pests or diseases to spread efficiently. It can make mechanised management easier, not impossible.

3 · Part 1

Core · Food supply · [1]

Answer: A

Explanation

High stocking density can facilitate disease spread. Intensive systems still require feed, produce waste and raise welfare considerations.

4 · Part 1

Core · Habitat destruction · [1]

Answer: C

Explanation

Biodiversity is the number of different species that live in an area. A counts individuals rather than species, B is the population size of one species, and D counts habitats rather than species.

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5 · Part 1 Core Habitat destruction [1]

Answer: D

Explanation

Clearing woodland removes habitat. Counting, seed banking and protection can support conservation rather than directly destroy it.

6 · Part 1 Core Habitat destruction [1]

Answer: B

Explanation

Tree roots hold soil particles together; once the trees are removed the loose soil is easily washed away by rain. A, C and D do not describe a mechanism that makes soil easier to wash away; in fact less evaporation from leaves leaves the soil wetter, not drier.

7 · Part 1 Core Pollution [1]

Answer: C

Explanation

Bacteria feeding on the sewage multiply and respire aerobically, using up the dissolved oxygen. Photosynthesis by algae adds oxygen rather than removing it, anaerobic respiration by fish is a consequence of low oxygen rather than its cause, and oxygen is not lost from a river by evaporation in this way.

8 · Part 1 Core Pollution [1]

Answer: A

Explanation

Non-biodegradable plastic persists and may cause ingestion or entanglement. It is neither necessarily soluble nor harmless food.

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9 · Part 1 Core Pollution [1]

Answer: B

Explanation

Methane and carbon dioxide absorb outgoing infrared radiation and contribute to warming. The other listed pairs are not the principal greenhouse gases described here.

10 · Part 1 Core Conservation [1]

Answer: C

Explanation

A sustainable resource is produced as rapidly as it is removed, so it does not run out. The other practices deplete the resource.

11 · Part 1 Core Conservation [1]

Answer: D

Explanation

Seed banks preserve viable plant material for future propagation. They complement habitat protection rather than require habitat removal or destruction of seeds.

12 · Part 1 Supplement Conservation [1]

Answer: A

Explanation

Larger mesh allows small fish to escape, grow and potentially breed, helping replenish stocks. It does not eliminate all predators or stop reproduction.

13 · Part 1 Supplement Conservation [1]

Answer: C

Explanation

Replanting helps replace harvested trees. The other measures reduce forest regeneration or convert land to another use.

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14 · Part 1 Supplement Conservation [1]

Answer: A

Explanation

Artificial insemination introduces sperm into the female reproductive tract and can facilitate planned crosses. It is sexual reproduction, not cloning.

15 · Part 1 Supplement Conservation [1]

Answer: B

Explanation

Low variation can leave few resistant or tolerant variants when conditions change. It does not guarantee universal adaptation or eliminate disease.

16 · Part 1 Core Food supply [4]

Mark scheme

(a) **[2]**

- Machinery allows larger areas to be cultivated or harvested efficiently. (1)

ALLOW ploughing, sowing or harvesting done faster / with fewer workers

- Fertiliser supplies mineral ions needed for plant growth. (1)

ALLOW named ions such as nitrate, phosphate, potassium / makes plants grow larger so greater yield

(b) **[2]**

- Insecticide reduces damage by insect pests. (1)

ALLOW kills insects that eat the crop / improves quality as well as yield

- Herbicide reduces competition with weeds for resources. (1)

ALLOW competition for light, water, mineral ions or space

Worked solution

(a) Machinery allows larger areas to be cultivated or harvested efficiently. Fertiliser supplies mineral ions needed for plant growth.

(b) Insecticide reduces damage by insect pests. Herbicide reduces competition with weeds for resources.

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17 · Part 1 Core Food supply [4]

Mark scheme

(a) **[2]**

- A single crop can be managed and harvested efficiently with machinery. (1)

> **ALLOW** any one advantage: all plants have the same requirements / higher yield per area / lower cost / easier to apply fertiliser or pesticide

- A pest or disease of that crop can spread through many plants. (1)

> **ALLOW** any one disadvantage: reduced biodiversity / loss of habitats / soil mineral ions depleted / more fertiliser or pesticide needed

(b) **[2]**

- Controlled conditions and reduced movement can increase production efficiency. (1)

> **ALLOW** any one advantage: more animals kept in a small area / animals grow faster because less energy is lost in movement or keeping warm / easier to control feeding and disease / cheaper meat, milk or eggs

- Crowding can increase disease spread or cause welfare problems. (1)

> **ALLOW** any one disadvantage: large amounts of waste can pollute water / antibiotics used routinely / high cost of feed and housing / land used to grow animal feed

Worked solution

(a) A single crop can be managed and harvested efficiently with machinery. A pest or disease of that crop can spread through many plants.

(b) Controlled conditions and reduced movement can increase production efficiency. Crowding can increase disease spread or cause welfare problems.

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18 · Part 1 Core Habitat destruction [4]

Mark scheme

(a) **[2]**

- There are fewer suitable breeding sites. (1)
- Reduced reproduction can cause the population to decline. (1)

ALLOW the species may become endangered or extinct locally

(b) **[2]**

- Predators of the fish may lose an important food source. (1)

ALLOW predator numbers fall or predators switch to other prey

- Organisms eaten by the fish may increase when predation is reduced. (1)

ALLOW competitors of the fish increase / effects pass further along the food chain

Worked solution

(a) There are fewer suitable breeding sites. Reduced reproduction can cause the population to decline.

(b) Predators of the fish may lose an important food source. Organisms eaten by the fish may increase when predation is reduced.

19 · Part 1 Core Habitat destruction [4]

Mark scheme

(a) **[2]**

- Roots no longer bind the soil as effectively. (1)
- Reduced interception and uptake allow more rapid surface runoff towards rivers. (1)

(b) **[2]**

- Fewer trees remove carbon dioxide by photosynthesis. (1)
- Burning or decomposition of cleared wood releases carbon dioxide. (1)

Worked solution

(a) Roots no longer bind the soil as effectively. Reduced interception and uptake allow more rapid surface runoff towards rivers.

(b) Fewer trees remove carbon dioxide by photosynthesis. Burning or decomposition of cleared wood releases carbon dioxide.

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20 · Part 1 Core Habitat destruction [4]

Mark scheme

(a) **[2]**

- The decrease is 48 − 18 = 30 species. (1)
- Biodiversity has decreased because fewer different species remain. (1)

(b) **[2]**

- A species may exist only in that habitat. (1)

ALLOW endemic species / species with a very small range

- If all its individuals lose the conditions needed to survive and reproduce, none may remain anywhere. (1)

ALLOW no other populations exist elsewhere to replace it

Worked solution

(a) The decrease is 48 − 18 = 30 species. Biodiversity has decreased because fewer different species remain.

(b) A species may exist only in that habitat. If all its individuals lose the conditions needed to survive and reproduce, none may remain anywhere.

21 · Part 1 Supplement Pollution [4]

Mark scheme

(a) **[2]**

- Extra nitrate and other ions increase producer growth. (1)
- When more algae die, decomposers have more dead material to break down. (1)

(b) **[2]**

- Aerobic respiration by decomposers uses dissolved oxygen. (1)
- Low dissolved oxygen prevents fish obtaining enough oxygen for respiration. (1)

Worked solution

(a) Extra nitrate and other ions increase producer growth. When more algae die, decomposers have more dead material to break down.

(b) Aerobic respiration by decomposers uses dissolved oxygen. Low dissolved oxygen prevents fish obtaining enough oxygen for respiration.

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22 · Part 1 Core Pollution [4]

Mark scheme

(a) **[2]**

- Animals may become entangled, restricting movement or causing injury. (1)
- Ingested plastic may block the gut or reduce feeding. (1)

> **ALLOW** any two: suffocation / toxic chemicals released from plastic / microplastics passed along food chains / habitat smothered by plastic

(b) **[2]**

- Decomposers do not readily break it down. (1)
- Water or wind can transport persistent plastic to other habitats. (1)

Worked solution

(a) Animals may become entangled, restricting movement or causing injury. Ingested plastic may block the gut or reduce feeding.

(b) Decomposers do not readily break it down. Water or wind can transport persistent plastic to other habitats.

23 · Part 1 Core Pollution [4]

Mark scheme

(a) **[2]**

- Fossil-fuel combustion releases carbon dioxide. (1)
- Cattle farming releases methane. (1)

(b) **[2]**

- They absorb more outgoing infrared radiation and reduce heat loss to space. (1)
- The enhanced greenhouse effect can raise global temperatures and alter climate. (1)

Worked solution

(a) Fossil-fuel combustion releases carbon dioxide. Cattle farming releases methane.

(b) They absorb more outgoing infrared radiation and reduce heat loss to space. The enhanced greenhouse effect can raise global temperatures and alter climate.

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24 · Part 1 Supplement Conservation [4]

Mark scheme

(a) **[2]**

- They are replenished by growth or reproduction. (1)
- Their removal must not exceed their replenishment over time. (1)

(b) **[2]**

- Replant or regenerate harvested forest. (1)

ALLOW felling quotas / protected areas of forest / education

- Limit fish catches or protect breeding periods. (1)

ALLOW catch quotas / closed seasons / controlled net type or larger mesh size / protected areas / monitoring

Worked solution

(a) They are replenished by growth or reproduction. Their removal must not exceed their replenishment over time.

(b) Replant or regenerate harvested forest. Limit fish catches or protect breeding periods.

25 · Part 1 Core Conservation [4]

Mark scheme

(a) **[2]**

- The predator increases mortality of birds or their young. (1)
- If deaths exceed successful reproduction, the population decreases. (1)

(b) **[2]**

- Control the introduced predator and protect the bird's habitat. (1)
- Use monitored captive breeding or protected breeding sites to increase survival and reproduction. (1)

ALLOW any two of: monitoring the population / protecting nesting sites / removing the predator / captive breeding and release / educating local people

Worked solution

(a) The predator increases mortality of birds or their young. If deaths exceed successful reproduction, the population decreases.

(b) Control the introduced predator and protect the bird's habitat. Use monitored captive breeding or protected breeding sites to increase survival and reproduction.

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26 · Part 1 Supplement Conservation [4]

Mark scheme

(a) **[2]**

- Quotas limit the rate of tree removal. (1)
- Replanting replaces harvested trees as the forest grows. (1)

(b) **[2]**

- Protected areas maintain habitats where logging is restricted. (1)
- Education encourages people to understand and support sustainable use. (1)

Worked solution

(a) Quotas limit the rate of tree removal. Replanting replaces harvested trees as the forest grows.

(b) Protected areas maintain habitats where logging is restricted. Education encourages people to understand and support sustainable use.

27 · Part 1 Supplement Conservation [4]

Mark scheme

(a) **[2]**

- Closed seasons allow fish to reproduce with less fishing pressure. (1)
- Quotas limit the number or mass removed from the stock. (1)

(b) **[2]**

- Monitoring shows whether stocks recover and whether limits need adjustment. (1)
- Suitable larger mesh lets more immature fish escape and later breed. (1)

Worked solution

(a) Closed seasons allow fish to reproduce with less fishing pressure. Quotas limit the number or mass removed from the stock.

(b) Monitoring shows whether stocks recover and whether limits need adjustment. Suitable larger mesh lets more immature fish escape and later breed.

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28 · Part 1 Supplement Conservation [4]

Mark scheme

(a) **[2]**

- It reduces the risk of species becoming extinct. (1)
- It retains genetic resources that may provide useful food, drugs or other products. (1)

> **ALLOW** any two of: maintains or increases biodiversity / protects a vulnerable ecosystem / maintains ecosystem functions such as nutrient cycling

(b) **[2]**

- It helps maintain nutrient cycling by interacting organisms. (1)
- It protects resource provision and the relationships supporting the community. (1)

> **ALLOW** named resources such as food, drugs, fuel or genes / habitat for many species / food webs kept intact

Worked solution

(a) It reduces the risk of species becoming extinct. It retains genetic resources that may provide useful food, drugs or other products.

(b) It helps maintain nutrient cycling by interacting organisms. It protects resource provision and the relationships supporting the community.

29 · Part 1 Supplement Conservation [4]

Mark scheme

(a) **[2]**

- Sperm from a selected male can be collected and transferred to another site. (1)
- The sperm can be introduced into a selected female's reproductive tract. (1)

(b) **[2]**

- Eggs are fertilised with sperm outside the female's body. (1)
- The resulting embryos are transferred to a female for development. (1)

Worked solution

(a) Sperm from a selected male can be collected and transferred to another site. The sperm can be introduced into a selected female's reproductive tract.

(b) Eggs are fertilised with sperm outside the female's body. The resulting embryos are transferred to a female for development.

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30 · Part 1 Supplement Conservation [4]

Mark scheme

(a) **[2]**

- There may be few alleles conferring resistance to a new disease. (1)
- There may be few variants suited to a changed climate or habitat. (1)

> **ALLOW** inbreeding makes harmful recessive alleles more likely to be expressed / population less able to adapt by natural selection

(b) **[2]**

- It preserves a wider range of inherited features. (1)
- This improves the chance that some offspring can survive future environmental challenges. (1)

Worked solution

(a) There may be few alleles conferring resistance to a new disease. There may be few variants suited to a changed climate or habitat.

(b) It preserves a wider range of inherited features. This improves the chance that some offspring can survive future environmental challenges.

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